

Dataset Scenarios Overview

Peking Opera Dynasty: 6 Generations of Lineage

Generation	Actor Name	Role	Artistic Style	Historical Period	Signature Role
1st	Chen Shou	Lead Actor	Traditional	18th Century	Chen Shou
2nd	Li Xiang	Lead Actor	Traditional	19th Century	Li Xiang
3rd	Wang Wei	Lead Actor	Traditional	20th Century	Wang Wei
4th	Zhang Li	Lead Actor	Traditional	21st Century	Zhang Li
5th	Liu Mei	Lead Actor	Traditional	22nd Century	Liu Mei
6th	Chen Shou	Lead Actor	Traditional	23rd Century	Chen Shou

Long Documents

Text/Meeting Query

Meeting Minutes

Natural Conversations

1. Curriculum-driven Rejection Sampling Fine-tuning with Knowledge Augmentation (CRSFT)

Illustration of Hardness Determination

Generate 2 SVGs

Judge

0 pass: Hard

1 pass: Medium

2 pass: Simple

Staged Curriculum Learning

Stage 3: Advanced
85% Hard + 15% Med
Complex Reasoning

Stage 2: Generalization
85% Med + 15% Simple
Diversity

Stage 1: Basic
100% Simple Data
Basic Concepts

Generator

Design Constraints

- No Overlap...
- No Miswiring...
- No Text Overflow...
- No ViewBox Clipping...
- No Misalignment...
- No Hallucination...

Knowledge Priors

2. Reinforcement Learning from Dual Rendering Feedback (RLDRF)

GRPO Update

Group Sampling

$S_1, S_2, \dots, S_G$

Generator

Composite Reward

A. Rule Verifier (Hard Constraints)
Format, Complexity
 $R_{complex}$ Score

B. Visual Verifier (Soft Constraints)
High-Tier Referee
 R_{visual} Score

Sum R_i

$R_1, R_2, \dots, R_G$

3. Generate-Verify-Repair Agent Loop (VRL)

Generate Agent

SVG

Render

Iterative Loop

Verify Agent

fail

pass

Error Log

Problem Attribute: ...
Suggestion: ...
Score: ...

Repair Agent

Standards

- Overlap
- Layout
- Canvas
- Connection
- Overflow
- Alignment

Final Output

ANNUAL PARTY STRATEGY

1. Budgeting

2. Logistics

3. Content

CONFIRM INPUTS

LOCK LOGISTICS

FORM TEAM